VLSI DSP Term Project option 3

Linear Regression module design

1. Background

Linear regression is one of the most important tools in statistical data analysis. It is used to determine the statistical relation between a dependent variable and an independent one, assuming that this relation can be modeled by a line. Assume X is an independent variable and Y is a dependent variable, linear regression finds the linear relation between two variables by the following formula

$$Y = \beta_0 + \beta_1 \cdot X \tag{1}$$

Given two data sequences $\{x_i \mid i = 1 \sim N\}$ and $\{y_i \mid i = 1 \sim N\}$, the two parameters of the linear regression, subject to the criterion of minimum mean square error, can be estimated as

$$\beta_1 = \frac{N\sum_{i=1}^{N} x_i \cdot y_i - \sum_{i=1}^{N} x_i \sum_{i=1}^{N} y_i}{N\sum_{i=1}^{N} x_i^2 - \left(\sum_{i=1}^{N} x_i\right)^2} \tag{2}$$

$$\beta_0 = \frac{\sum_{i=1}^{N} y_i - \beta_1 \sum_{i=1}^{N} x_i}{N} \tag{3}$$

The mean square error MSE can be calculated as

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \beta_0 - \beta_1 x_i\right)^2 \tag{4}$$

2. Design features

Given two input data sequences $\{x_i \mid i = 1 \sim N\}$ and $\{y_i \mid i = 1 \sim N\}$, the linear regression module calculates the two parameters β_0, β_1 and the MSE value. The hardware design should support the following features

- Capability of processing a continuous data flow, i.e., β_0, β_1 and MSE should be computed on the fly along with the input data streams. At the end of the data streams, the computations should be completed as well (a constant number of clock cycle delay, e.g. m clock cycles and $m << N$, is acceptable)
- Capability of admitting any sequence length N, an end-of-sequence (EOS) symbol will be

placed at the end to indicate the end of the sequence

- Free of any line buffer or memory, just a few registers to hold the intermediate results are enough to complete the computations

Assume the maximum length of the data sequence is 256 and the word length of every input data is 16 bits.

3. Design considerations

To accomplish the aforementioned design features, several design issues must be resolved.

- The equations (2) ~ (4) should be reformulated so that they can be computed incrementally or on-the-fly with the advance of input data.
- For β_1, this is easy as the accumulation (or the summation) can be performed incrementally
- For β_0, although its value depends on the final outcome of β_1, the two summation terms can still be computed incrementally. However, the completion of computation would be deferred until the availability of β_1.
- For MSE calculation, this is the most challenging part of this assignment. Its calculation depends on the final values of β_0 and β_1, and the dependence associated with every y_i from i = 1 to N. Therefore, some kind of ***look ahead*** transform must be applied so that the calculation of every prediction error term $(y_i - \beta_0 - \beta_1 x_i)^2$ proceeds with partial results of β_0 and β_1. Once this is resolved, you can achieve a design free of any line buffer or memory to hold the values of two data sequences.

4. Hand in requirements

The following items are required

- Algorithm derivation process (or reformulated equations) to show how the computing concurrency can be achieved
- Circuit diagram of the derived design
- Verilog coding and verification results
- Synthesis results using TSMC ADFP process technology (default)
- FPGA implementation results (this option is open to undergraduate students only), the target device is Xilinx Virtex-6 XC6VSX475T -2 FF1156. Please report the maximum working frequency, the number of used slices, and the latency (the time elapse between

the end of the input sequence and the outcome of the linear regression parameters and MSE). For your reference, the numbers reported in the paper are 238MHz, 484 slices and 38 ns, respectively.